Diophantine tuples from cyclic square patterns

Human Compression project: F011 application

This paper recovers the principal bounds of Kim–Yip–Yoo (2024), with explicit corrections to intermediate statements and arguments. It uses the complete companion *Square patterns and cyclotomic degree bounds over finite fields*. A Diophantine tuple consists of distinct nonzero field elements whose products in distinct positions become squares after adding one. Zero is allowed as the resulting square, not as a tuple element. All fields have odd characteristic p. All logarithms without a base are natural.

For a square field order $q = t^2$, its multiplicative subfield group $\mathbb{F}_t^*$ consists of squares in $\mathbb{F}_q$: if $a \in \mathbb{F}_t^*$ then $a^{(q-1)/2} = (a^{t-1})^{(t+1)/2} = 1$. Products plus one of subfield elements remain in the subfield, and zero is allowed. Thus the full nonzero subfield is a tuple of size $\sqrt{q}-1$, as recorded in the source introduction. Its cited bound $M(q) \le \sqrt{q} + 5/2$ for all odd q is a retained background result.

1 The principal lower bound

Proposition 1 (Consecutive powers)**.** *Let $m \ge 2$ and let y be a square of order at least m. Define the integer interval*

$$J_m = \{-\lfloor m/2 \rfloor, \dots, \lceil m/2 \rceil - 1\}.$$

If $q \equiv 1 \pmod 4$, $r^2 = -1$, and $y^j - 1$ is a square for every $1 \le j < m$, then $\{ry^i : i \in J_m\}$ is a Diophantine m-tuple. If 2 is a square and every $y^j + 1$, $1 \le j < m$, is a square, then $\{y^i : i \in J_m\}$ is a Diophantine m-tuple.

Proof. The interval has m elements with pairwise differences at most $m-1$, so their powers are distinct. Sums of two distinct indices have absolute value at most $m-1$. In the first case the required expression is $1 - y^{i+j}$. For positive sums it is the product of two squares, -1 and $y^{i+j} - 1$. At zero it is zero. For a negative sum $-j$, it is $(y^j - 1)/y^j$, again a square. In the second case the positive sums are covered by assumption, the zero sum gives 2, and a negative sum gives $(y^j + 1)/y^j$. All tuple entries are nonzero. □

Theorem 2 (Source Theorem 1.1, with a repaired proof)**.** *Let $q > 7$ be a power of the odd prime p, with $q \equiv 1, 5, 7 \pmod 8$. Then*

$$M(q) \ge \left\lfloor \frac{p}{p-1} \left(\frac{\frac{1}{2}\log q - 2\log\log q}{\log 2} + 1 \right) \right\rfloor.$$

For fixed p, along these congruence classes this is $M(q) \ge (p/(p-1) - o(1)) \log_4 q$.

Proof. Let m be the displayed integer. If $m \le 1$ the singleton tuple suffices. Otherwise the companion parameter lemma and square-pattern proposition give y of order at least m. For $q \equiv 1 \pmod 4$ use its minus pattern and a square root of -1. For $q \equiv 7 \pmod 8$, use its plus pattern. Here 2 is a square: the extension degree is odd, its prime has residue seven modulo eight, and the usual quadratic-character supplement gives this assertion in the prime field, hence in the extension. Proposition 1 applies. Dividing the expression by $\log_4 q$ gives the asymptotic form because $\log\log q / \log q \to 0$. □

The exact same proof gives source Corollary 3.2, without requiring a newly acquired proof of Cohen's theorem. It retains the classical Weil bound, precisely as stated in the companion. This is a reworking of the source method; it claims no historical originality. The corollary's only additional odd prime power $q \equiv 1 \pmod 4$ with $q \leq 7$ is $q = 5$; there its displayed bound is at most one, so a singleton suffices.

Proposition 3 (The remaining residue class)**.** *For $q > 7$ with $q \equiv 3 \pmod 8$, let m be the integer in Theorem 2. If $m \geq 2$, there is a Diophantine tuple of size $\lfloor m/2 \rfloor + 1$. Consequently*

$$M(q) \geq \left(\frac{p}{2(p-1)} - o(1) \right) \log_4 q$$

for fixed p along this residue class.

Proof. The companion plus-pattern proposition does not require 2 to be a square. Choose the resulting y and use the exponents $0, 1, \ldots, h-1$, where $h = \lfloor m/2 \rfloor + 1 \leq m$. Two distinct exponents have sum between one and $2h - 3 \leq m - 1$. Thus no product tests the nonsquare value 2. Distinctness follows from the order bound. This supplies the full asymptotic assertion of source Remark 3.6 and does not assert maximality. □

2 The stronger bound in residue class seven

Proposition 4 (Centered half-integer powers)**.** *Let y be a nonzero square of order at least $n \geq 2$, and let $z^2 = y$. If 2 and all $y^j + 1$, $1 \leq j \leq n-2$, are squares, then*

$$\{z^{-(n-1)}, z^{-(n-3)}, \ldots, z^{n-3}, z^{n-1}\} \tag{1}$$

is a Diophantine n-tuple.

Proof. Write its entries as $z^{2i-(n-1)}$, $0 \leq i < n$. The ratio of two entries is y^{i-j}, which cannot be one for $0 < |i - j| < n$. The product of distinct entries is $y^{i+j-(n-1)}$. Its exponent has absolute value at most $n - 2$, since the smallest distinct pair is $\{0, 1\}$ and the largest is $\{n-2, n-1\}$. A zero exponent gives 2; positive exponents are covered by hypothesis; and for a negative exponent $-h$ the expression is $(y^h + 1)/y^h$, a square. All entries are nonzero. For odd n, (1) is the usual centered integer progression in y. For even n, it uses half-integer powers through the chosen square root. This is why the largest necessary exponent is still $n - 2$. □

Theorem 5 (Source Theorem 3.5, with a repaired construction)**.** *If $q > 7$ and $q \equiv 7 \pmod 8$, then*

$$M(q) \geq \left\lfloor \frac{p}{p-1} \left(\frac{\frac{1}{2} \log q - 2 \log \log q}{\log 2} + 2 \right) \right\rfloor .$$

Proof. Let n be the displayed integer. For $n \geq 4$, the companion's stronger parameter lemma and shorter-pattern proposition give a square y of order at least n with all required $n - 2$ power conditions. The quadratic-character supplement gives that 2 is a square, so Proposition 4 applies.

For completeness, every eligible field has a tuple of size three, which handles all $n \leq 3$. Indeed

$$\#\{y \neq 0, 1 : \chi(y) = \chi(y+1) = 1\} = (q-7)/4 > 0. \tag{2}$$

To check the identity, the complete sum of $(1 + \chi(y))(1 + \chi(y+1))/4$ is $(q-1)/4$: the linear sums vanish and $\sum \chi(y(y+1)) = -1$. The latter follows by completing the square and counting $(u - v)(u + v) = c$ for nonzero c, which has $q - 1$ solutions. At zero the weight is $1/2$; at one it is one since 2 is square; and at -1 it is zero since -1 is nonsquare. Elsewhere the weight is the required indicator. This proves (2). Choose such a y. Squares have odd order because $(q-1)/2$ is odd, and $y \neq 1$, so its order is at least three. The set $\{y^{-1}, 1, y\}$ is therefore a tuple, either directly or by Proposition 4. □

3 Maximal tuples

Proposition 6 (Source Proposition 2.3)**.** *Every maximal Diophantine m-tuple satisfies $q < 4^{m-1}m^2$.*

Proof. For $m \geq 3$, let its entries be a_i and apply the companion prescribed- character lemma with $d = 2$, shifts a_i^{-1} and signs $\chi(a_i)$. Every resulting point makes all $a_i x + 1$ nonzero squares. Removing the at most $m + 1$ points in $\{0, a_1, \ldots, a_m\}$ leaves at least

$$\frac{q}{2^m} - \left(\frac{m}{2} - 1 + 2^{-m}\right)\sqrt{q} - \frac{3m}{2} - 1.$$

If $\sqrt{q} \geq 2^{m-1}m$, this exceeds zero: the difference between it and $q/2^m - m\sqrt{q}/2$ is at least $(2^{m-1} - \frac{1}{2})m - 3m/2 - 1 = (2^{m-1} - 2)m - 1 > 0$. It would extend the tuple, a contradiction.

For $m = 2$ and $q \geq 17$, use directly $W(x) = (1+\chi(ax+1))(1+\chi(bx+1))/4$ for distinct nonzero a, b. The two linear sums vanish and the quadratic sum is $-\chi(ab)$; hence $\sum W = (q - \chi(ab))/4 \geq 4$. The latter identity follows, for example, by completing the square: for nonzero c, counting solutions of $v^2 = u^2 - c$ through $(u - v)(u + v) = c$ gives $\sum_u \chi(u^2 - c) = -1$. Removing $0, a, b$ costs at most three, so an extending point remains. Since odd $q \geq 16$ means $q \geq 17$, this proves the required strict bound when $m = 2$.

A singleton is not maximal for $q \geq 5$: the weight $(1+\chi(ax+1))/2$ has total $q/2$, and removing $0, a$ costs at most two. For $q = 3$ the two nonzero elements form a tuple. The empty tuple is never maximal. This covers every possible m. □

The asymptotic bound for all maximal tuples follows: for any fixed $\epsilon > 0$, $m \leq (1 - \epsilon)\log_4 q$ would make $4^{m-1}m^2 < q$ for sufficiently large q.

4 Exact source corrections and retained scope

The displayed set in source Proposition 3.1 has $m + 1$ terms but assumes only $\mathrm{ord}(y) \geq m$. For example, over $\mathbb{F}_{17}$ take $m = 2$, $y = -1$ and $r = 4$. Both $y - 1 = 15$ and $y^2 - 1 = 0$ are squares, and y is a square, but the displayed list is $(13, 4, 13)$ and has only two distinct elements. The set is still Diophantine. Its claimed cardinality is false. With the strict order hypothesis $\mathrm{ord}(y) > m$ the original proof gives the stated $m + 1$ cardinality. Proposition 1 instead uses exactly m consecutive powers under the original weak order bound, which is enough for the unchanged main lower bound.

Source Proposition 3.3 is not subject to that collision in its stated section $q \equiv 7 \pmod 8$: squares have odd order, so an even m and $\mathrm{ord}(y) \geq m$ imply $\mathrm{ord}(y) > m$. Its cyclotomic proof is valid when the displayed quotient is interpreted as the polynomial product identity.

The introductory least-nonresidue example includes zero despite the paper's nonzero-element convention. A valid corresponding set is $\{1, \ldots, \lfloor\sqrt{n(p)}\rfloor\}$: products at distinct positions plus one are less than $n(p)$, hence squares. The literal bound $M(p) \geq \sqrt{n(p)} + 1$ is false for $p = 3$, where $n(3) = 2$ but $M(3) = 2$. This correction leaves its stated asymptotic comparison unchanged.

The source counting proof has additional issues which the companion avoids: the degree-one factor $\Phi_0 = X$ must be counted, and the removal of zero must be paid. The companion deliberately uses a weighted filter in general; in the source's $q \equiv 7 \pmod 8$ plus pattern its nonzero-point filter is an actual indicator, since a square has odd order and cannot be a root of any Φ_{2i} with $p \nmid i$. Source Proposition 2.3 similarly omits one point in its displayed exclusion estimate; the proof above includes it and handles the small tuple sizes separately.

Source Theorem 3.5, in the section $q \equiv 7 \pmod 8$, replaces the final $+1$ in Theorem 2 by $+2$. This is a separate, stronger claim. Its proof asserts $T \leq B + 2$ for $m = \lfloor\alpha(B + 2)\rfloor$, which does not follow and can fail. For $q = 7^{11}$ the prescribed integer is $m = 10$, giving $T = 9$, whereas

$8.6006 < B + 2 < 8.6008$. There is also an exact small-parameter contradiction to the displayed character-count lower bound. For $q \equiv 7 \pmod 8$ and $m = 2$, its count is

$$N(2) = \#\{y \neq 0, 1 : \chi(y) = \chi(y+1) = 1\} = (q-7)/4.$$

Indeed the complete weight sum is $(q-1)/4$ by the elementary quadratic sum evaluated in the preceding proof; zero has weight $1/2$, one has weight one, and -1 has weight zero. The source instead gives $N(2) \geq (q + \sqrt{q})/4 - 1$, since $T = 2$ and $S = 1$, which is larger than the exact count. The value $q = 7^5$ lies in the theorem's branch and gives its prescribed $m = 2$, so this is an error inside the proof's actual parameter range. This refutes that intermediate estimate, not the claimed existence of a tuple. Its final logarithmic deduction also does not follow merely from $\alpha < 2$. These defects in the printed route do not refute the existence theorem. Theorem 5 proves the exact stronger bound using centered half-integer powers, the shorter cyclotomic filter, an explicitly stronger order exclusion, and the corrected parameter bound $T \leq B + 2 + 1/p$. All these steps are written in full in the two papers.

The source conjecture $M(q) \geq (1 - o(1)) \log_2 q$ is recorded without an answer or a current-status claim. The cited square-field upper bound, integer-tuple history and GRH least-nonresidue estimates remain attributed background, not fresh theorems proved here. No new Lean or numerical compression measurement is supplied.